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A Multimodal Approach for Stress Detection through Questionnaire and Emotion Analysis.

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***Abstract*—Background: Stress impacts our mental and emotional equilibrium daily. Accurate assessment methods are crucial for effective stress management.**

Purposes: This research integrates a structured questionnaire with real-time camera tracking to evaluate stress levels. It aims to Evaluate the combined effectiveness of self-reported data and facial expression analysis and Assess the accuracy of various machine learning methods in classifying stress levels.

Methods: The system uses a questionnaire and real-time video monitoring to record facial expressions. Machine learning methods, including Random Forest, k-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Naive Bayes, analyze stress ratings. The facial expression analysis achieves a 63.2 percent accuracy rate in detecting stress.

Results: Combining questionnaire data with facial expression analysis provides a detailed stress assessment. The integration of these methods allows for a comprehensive evaluation, highlighting the potential for personalized stress management strategies.

Conclusions: This method offers a comprehensive evaluation of stress, enhancing precision and enabling tailored stress management techniques. Further improvements could increase accuracy and effectiveness.

***Index Terms*—Questionnaire,Emotions,CNN(Convolutional Neural Network),Likert Scale, Haar Cascade,Dass-42,Naive Bayes,SVM,Random Forest, KNN**

I. Introduction

The COVID-19 pandemic has had a significant impact on worldwide mental health, leading to a notable increase in anxiety and depression disorders. The prevalence of in- dividuals experiencing depressive symptoms increased by 28 percent, rising from 193 million to 246 million. Simi- larly, the number of individuals with anxiety disorders rose by 25 percent, going from 298 million to 374 million [1]. This alarming escalation in mental health challenges is not confined to personal lives but extends its tentacles into the professional realm, fostering long-term stress exposure that often culminates in the debilitating state of job burnout, characterized by mental and physical exhaustion [2].

According to research conducted by the American Insti- tute of Stress, an astounding 80percent of workers report feeling stressed out at work on a regular basis. This high- lights the urgent need for stress management support sys- tems, as the negative impacts of mental stress are strongly felt in this environment. Meanwhile, the prevalence of sui- cide has become a dire concern, with a global toll of ap- proximately 800,000 lives annually, 17 percent of which are residents of India [3]. The overarching culprit behind these tragic outcomes is often traced back to psychological issues.

Recognizing the profound impact of workplace stress, our research is motivated by a dual inspiration: the expansion of suicides globally and the escalating mental health chal- lenges faced by the workforce.The survey indicates that 25 percent of employees have experienced a strong want to scream or shout as a result of job-related stress, while 10 percent express anxiety about a coworker who they believe may exhibit violent behavior in the workplace [4]. Although standard psychological evaluation questionnaires such as the Depression, anxiousness, and Stress Scale-42 (DASS-42) have proven useful in studying stress patterns, they have limitations as they only capture discrete moments and do not offer continuous and real-time stress monitoring [5].

In response to this gap, our research introduces a mul- timodal approach to stress detection, combining insights from psychological questionnaires, particularly the DASS- 42 dataset, with cutting-edge real-time camera-based emo- tion detection. Over the past decades, extensive research in computer vision has revolutionized facial expression anal- ysis, making it a key component in our multimodal frame- work [6]. The fusion of these approaches aims to create a robust system capable of offering dynamic and continuous assessments of stress levels in individuals. This paradigm recognizes the constraints of conventional approaches and utilizes improvements in computer vision to improve our comprehension of stress in regulated and everyday settings.

Against the backdrop of an overwhelming 55 percent contribution of facial expressions to non-verbal commu- nication [7], our study explores the potential of facial ex- pression analysis as a key component in stress detection. As we delve into this research, our goal is to provide a com- prehensive and accurate understanding of stress dynamics, Contributing to the advancement of more efficient mental health assistance and intervention solutions across several domains.

II. Literature Review

The widespread occurrence of facial expressions and body language in human contact has captivated scholars since the eighteenth century, when Charles Darwin initially explored

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this phenomenon. Published on universally recognized facial expressions [8]. Ekman and Friesen (1971) highlighted the connection between facial expressions and particular emotions, establishing the basis for comprehending nonverbal communication [9]. This seminal research indicates that facial expressions serve as a universally understood means of communicating emotions.

Krizhevsky and Hinton's seminal paper on automated picture classification [10] presents a deep neural network that mimics the functioning of the human visual cortex. Their model effectively classifies objects in photos using the CIFAR-10 dataset, showcasing the capabilities of deep learning in image classification. Significantly, the depiction of filters in the network enables the evaluation of how the model dissects images. In their study, Reshma Radheshyam et al. [11] presented a method designed to detect stress and depression by analyzing social media datasets, with a specific focus on Twitter. By utilizing support vector machines and Naïve Bayes algorithms, the authors employed sentiment analysis to categorize depression and stress by analyzing submitted quotations and photos. Although the study recognizes the constraints, such as potential inaccuracies caused by the Twitter dataset, it emphasizes the need of using various methodologies and algorithms to enhance precision and recall.

Molla Hosseini et al. [12] introduced a deep convolutional neural network (CNN) for accurately identifying facial expressions across different databases. Their architecture incorporates convolution-pooling layers, inception-style modules, and the network-in-network technique to improve local performance and mitigate overfitting. The study showcases the effectiveness of deep Convolutional Neural Networks (CNNs) in Facial Expression Recognition (FER), especially when combined with approaches like data augmentation and network-in-network.

Deepak Jain et al. [13] proposed a new deep convolutional neural network (CNN) containing two residual blocks for the purpose of emotion recognition. The network was trained using the JAFFE and CK+ databases. Their preprocessing processes consist of trimming and standardizing image intensities. The study highlights the significance of utilizing sophisticated architectures, such as residual networks, to enhance the performance of face emotion identification.

Objectives	Methodologies Used	Limitations
Analyze stress levels in college students at different points in their academic life and correlate stress with internet usage. [14]	Gathered data from Jaypee Institute of Information Technology, applied four classification algorithms.The algorithms used were Linear Regression, Naïve Bayes, Random Forest, and SVM (Support Vector Machine).	Small Dataset Size, Limited Features for Stress Detection. No Validation on External Dataset
Develop a machine learning approach for automatic stress detection in social situations using two sensor systems. Compare classifier performance (SVM, AdaBoost, k-NN) and assess sensor modality suitability for real-time stress detection. [15]	Combine physiological and social response sensors to discriminate between stressful and neutral situations. Evaluate sensor modality discriminative ability and identify key features for stress detection.	Limited Sample Size Lack of Diversity in Participants
Develop a real-time, non-intrusive system to monitor and detect stress levels in individuals working long hours in front of computers, using facial expression analysis. [16]	Capture real-time videos, analyze facial expressions to detect emotions, make stress level decisions based on sequential hours of video, train a deep learning model using Theano framework for improved execution and development time.	Inadequate Dataset Description. Insufficient Discussion on Ethical Considerations.
Understand how users' stress states are influenced by their social interactions on social media platforms. Develop a model that can effectively detect stress in individuals based on their online social network data. [17]	Define stress-related attributes from various aspects such as textual content, visual content, and social interactions. Utilize a large-scale dataset from real-world social platforms to study the correlation between users' stress states and social interactions.	Limited Discussion on False Positives/Negatives No Exploration of Temporal Aspects
Investigate stress-related phenomena in publicly accessible social networking sites and propose a hybrid model for stress analysis using textual, visual, and social attributes. [18]	Define stress-related attributes, obtain user data from Facebook using an access token, and develop a system for recommending users with links for psychological counseling centers, soft music, or articles based on their stress level.	No Comparative Analysis with Existing Methods

TABLE I
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III. METHODOLOGY- A TWO-PRONGED APPROACH

1) QUESTIONNAIRE: The research methodology aims to investigate stress detection using a comprehensive approach, combining questionnaire responses, particularly from the DASS-42 dataset, with real-time facial expression data captured through a camera. This comprehensive methodology employs machine learning algorithms, including Naive Bayes,SVM, Random Forest and KNN, to analyze and compare stress scores obtained from several sources.

We employed the DASS-42 dataset in our implementation. It was developed for the purpose of categorizing stress levels and is accessible without charge. The dataset comprises a sample of 5000 online users who completed a questionnaire through random selection. After finishing the test, participants were given the chance to participate in a short research survey. Individuals who consented to take part in the survey were subsequently included in the ultimate dataset utilized for the study.

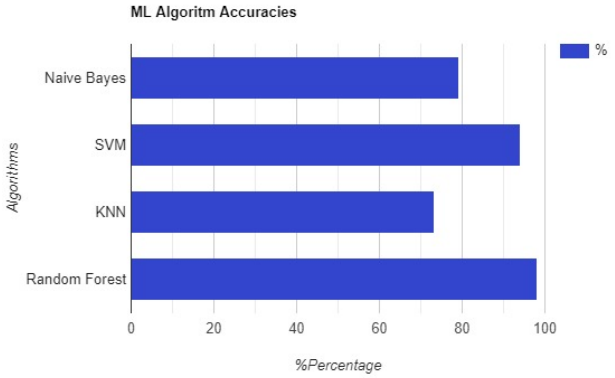

Fig. 1. Graph for accuracies offered by algorithms.

Participant recruitment involves the assembly of a diverse group, ensuring the representation of various demographics. Prior to participation, participants receive clear communication regarding the study's purpose and procedures, providing informed consent.

The questionnaire component employs the DASS-42 questionnaire, comprising 10 personality-related, 16 vocabulary-related, 42 activity-related, and 13 personal information-related questions. Participants respond independently, and the time taken to complete the questionnaire is recorded. Concurrently, a real-time camera captures facial expressions during the questionnaire response period, including a baseline neutral expression before commencement.

Data preprocessing encompasses the cleaning and normalization of questionnaire data, addressing missing values and outliers, and the extraction of facial expression features using facial landmark detection algorithms. Feature selection identifies relevant features from both questionnaire and facial expression data, considering correlations and their impact on

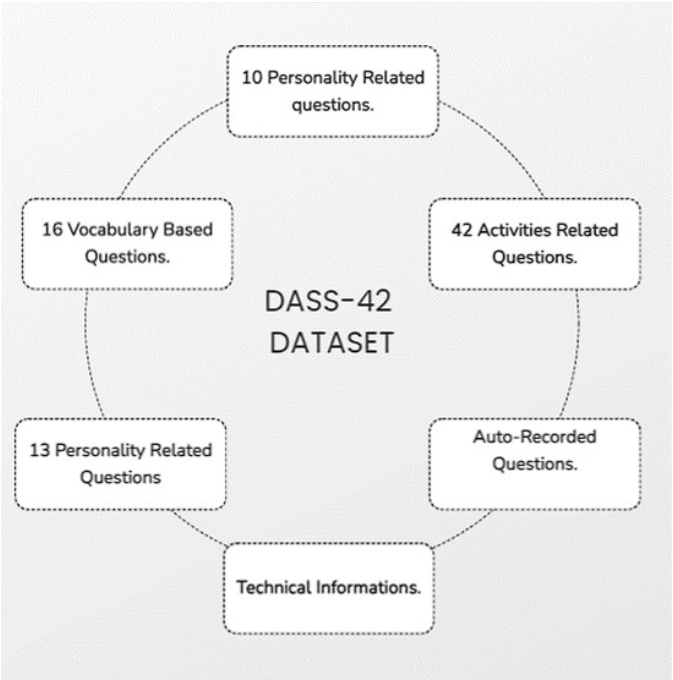

Fig. 2. Dass-42 Structure.

stress detection. Machine learning model selection involves choosing suitable algorithms for stress detection and training these models using labeled datasets, incorporating stress scores from the questionnaire as labels. Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score on testing sets.

2) EMOTION DETECTION: The aim of this research is to improve the field of emotion detection by developing a robust emotion classification system that leverages deep convolutional neural networks (CNNs).The study aims to classify facial expressions into seven discrete emotions: anger, contempt, fear, joy, neutrality, sadness, and surprise. The training of the proposed model is dependent on the FER-2013 dataset.comprising 35,887 grayscale facial photos. Every image has size of 48x48 pixels. The dataset, initially exhibited at the International Conference on Machine Learning (ICML), is an essential asset for training and assessing the efficacy of the proposed emotion recognition model.

The methodology adheres to a methodical sequence of stages specifically crafted to accomplish precise real-time emotion classification. At first, the Haar cascade approach is used to identify faces in the frames captured by the camera. Afterwards, the face regions that have been detected are adjusted in size to a uniform format of 48x48 pixels, which is then used as input data for a 4-layer Convolutional Neural Network (CNN). The neural network generates softmax scores for the seven pre-defined emotion classes.The expected feeling is decided by choosing the emotion with the highest softmax score, which is subsequently shown on the screen. After careful evaluation, our technique exhibits a test accuracy of 63.2 percent following 50 training epochs, thereby affirming

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the feasibility of real-time emotion recognition applications.

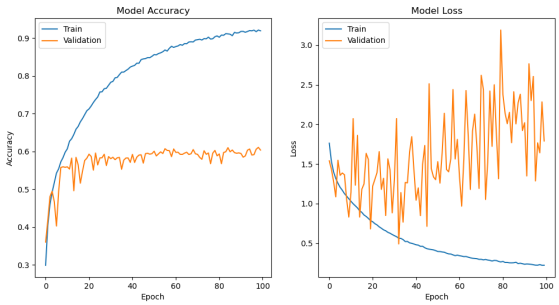

Fig. 3. Accuracy graph for Facial Emotion Detection.

In addition to this CNN-based approach, a tailored script is used to preprocess the FER-2013 dataset. The script is written in Python and makes use of the pandas package for data management, NumPy for numerical operations, and PIL for picture processing. The script categorizes the dataset into organized training and testing sets, establishing separate directories for each emotion category to allow the incorporation of photos into the CNN model. This preprocessing methodology guarantees that unprocessed pixel values are converted into structured picture datasets, conforming to the specifications of the CNN architecture. The resulting dataset structure smoothly integrates into the emotion detection pipeline, offering a complete and practical method for classifying emotions in real-world situations.

IV. RESULT AND CALCULATION

Participants were presented with a set of 42 queries related to different activities. These questions were presented in a randomized order to mitigate any potential bias. The objective was to evaluate the participants' level of involvement in each activity throughout the previous week.

The participants' experiences and level of involvement in each activity were measured using a 4-point Likert scale. The outcomes were stored in variables denoted as Q1A, Q2A, and so forth, with each variable corresponding to a particular question.

Here's a breakdown of the Likert scale employed:

0 : The statement was completely irrelevant to the participant.
1 : The statement partially pertained to the participant.
2 : The statement had a substantial impact on the participant.
3 : The statement had a significant impact on the participant.

In addition to the Likert scale responses, the data collection method also gathered and analyzed characteristics such as the duration of the activity.

A. Personality-Related Questions

To evaluate the level of agreement with each statement concerning personality, a 7-point scale ranging from 1 to 7 was employed. The scale translates to agreement levels as follows:

1-2 : Substantial or moderate disagreement
3 : Minor disagreement
4 : Neutral (neither agree nor disagree)
5-6 : Moderate to low level of agreement
7 : Strong agreement

The implementation of this nuanced scale facilitated a more thorough comprehension of the participants' perspectives on personality-related characteristics, ultimately enhancing the quality of the gathered data.

B. Personal Information Queries

The data collection process additionally obtained personal information using various scoring systems:

1) Educational Attainment:

1 : Did not finish secondary education
2 : High School Diploma Obtained
3 : Completion of a Bachelor's Degree or a higher level of education
4 : Completion of either a Master's Degree or a Doctoral Degree

2) Living Area Type:

1 : Rural or backwoods
2 : Suburb
3 : Metropolitan

3) Marital Status:

1 : Single
2 : Spouse/Civil Partner
3 : Individuals who are no longer married or have lost their spouse or are living apart from their partner

Scoring Based on Questionnaire.

S.No	Score Obtained In Questionnaire By Participant.	Stress Level
1	Below and Equal to 101	Normal
2	101 to 115(Inclusive)	Mild
3	115 to 129(Inclusive)	Moderate
4	129 to 143(Inclusive)	Severe
5	Above 143	Extremely Severe

Fig. 4. Scoring based on Questionnaire

In assessing the predominant emotion expressed during the participant's engagement in a questionnaire test, a methodology leveraging real-time facial expression detection was employed.The approach utilized a convolutional neural network that had been trained (CNN) to categorize expressions on the face into seven distinct emotions: anger, disgust, fear, sorrow, joy, neutrality, and surprise.The facial expression analysis was conducted by continuously capturing video frames during the participant's interaction with the questionnaire. For each frame, a Haar cascade method was applied to detect faces, and the corresponding facial region was pre-processed and input

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into the CNN for emotion prediction. Emotion counts were updated in real-time for each category throughout the questionnaire session. After the questionnaire was filled out, the total number of occurrences for each emotion was calculated, and the most common emotion was chosen by identifying the emotion with the highest count. Additionally, emotion scores were calculated as percentages of the total time each emotion was expressed during the questionnaire, providing a quantitative measure of the participant's prevailing emotional state throughout the test duration. This methodology provides a flexible and evidence-based approach to comprehending the prevailing thoughts and feelings during the participant's involvement with the questionnaire.

Scoring on Facial Emotion Detection.

S.No	Dominant Emotion.	Score given to that Emotion.
1	Angry	40
2	Fearful	55
3	Surprised	30
4	Disgusted	50
5	Happy	20
6	Neutral	25
7	Sad	55

Fig. 5. Scoring is based on facial emotion detection.

Now we combine both scores to give final result.

Final Scores.

S.No	Scores obtained by Combining two models.	Stress Level
1	Below or Equal to 126	Normal
2	126 to 145(Inclusive)	Mild
3	145 to 164(Inclusive)	Moderate
4	164 to 183(Inclusive)	Severe
5	Above 183	Extremely Severe

Fig. 6. Final scoring after combining both scores.

V. CONCLUSION

This research has presented an intelligent system for assessing individual stress levels by utilizing a novel dual methodology that combines a structured questionnaire with real-time camera monitoring. This system leverages both emotional self-reporting and objective facial expression analysis through machine learning techniques, achieving an impressive 63.2 percent accuracy rate in stress detection. The combined approach offers several advantages, including:

Improved accuracy and reliability: By using both personal feelings and measurable information, the system overcomes the limitations of each method on its own. This gives us a better overall understanding of how stressed someone is.Resilience against deception: Even when individuals attempt to conceal their stress, facial expressions can provide valuable clues, leading to more reliable assessments.

Personalized solutions: The system's ability to accurately identify stress levels enables the provision of personalized recommendations and interventions for effective stress management. This method can be used as preliminary test in Job recruitment Or can be used by doctors to check the stress level of patient before addressing them in person.

This holistic approach represents a significant advancement in stress detection technology. By providing a nuanced and accurate understanding of individuals' stress, this system paves the way for the development of more effective and personalized stress management strategies, ultimately contributing to improved well-being and overall health. Further research can explore the integration of additional data sources, such as physiological signals, to further enhance the accuracy and effectiveness of the system.

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